

Deepfake Image Detection: A Review of Existing Methods and a Hybrid CNN-Based Proposed Framework

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Abstract - The rapid advancement of generative adversarial networks (GANs) and related synthesis techniques has enabled the creation of highly realistic deepfake images, posing significant risks to security, privacy, and trust in digital media. Existing detection methods range from handcrafted forensic features to deep learning-based models, yet they often face challenges such as dataset bias, limited cross-domain generalization, and vulnerability to adversarial manipulations. This paper reviews state-of-the-art methods for deepfake and image forgery detection, highlighting critical research gaps, particularly in robust feature representation and scalability. To address these limitations, we propose a hybrid convolutional neural network (CNN)-based framework that integrates block-based ResNet-50 for effective feature extraction and VGG-16 for classification. Publicly available datasets such as Celeb-DF and the DeepFake Detection Challenge (DFDC) are discussed alongside challenges of real-world deployment. The study concludes by outlining future research directions, including multimodal detection, continual learning, and explainable AI for improved interpretability.

Keywords - Deepfake detection, Image forensics, Convolutional neural networks, ResNet-50, VGG-16

1 Introduction

The proliferation of deepfakes, driven by advances in generative adversarial networks (GANs), poses significant risks to trust, privacy, and information integrity in digital media

[1]. Classical image forensics based on handcrafted cues (e.g., lighting/consistency checks, noise analysis, and compression artifacts) has struggled to keep pace with the realism and diversity of modern synthetic content [1], [4].

Deep learning has therefore become central to detection. Afchar et al. introduced MesoNet, a compact CNN tailored to mesoscopic artifacts in manipulated faces, demonstrating that appropriately designed architectures can capture subtle synthesis traces [3]. In parallel, Rossler et al. released FaceForensics++, a benchmark that established Xception-based detectors as strong baselines while also revealing sensitivity to dataset conditions and manipulation types [2]. Broader surveys continue to underscore persistent challenges especially cross-dataset generalization and resistance to evolving manipulation pipelines motivating hybrid and transfer-leveraged strategies to improve robustness and scalability [4]. This paper reviews the landscape and then proposes a hybrid CNN-based framework that exploits block-based ResNet-50 for robust feature extraction and VGG-16 for classification, designed to improve generalization and practical deployability for deepfake image detection.

2 Literature Review

The detection of manipulated visual content has been studied under both deepfake detection and general image forgery detection. While the former primarily focuses on GAN-based facial manipulations, the latter addresses a wider range of tampering such as copy-move or splicing. Insights from both domains are highly relevant to building robust deepfake detection frameworks.

Hosny et al. [5] developed an efficient CNN model for copy-move image forgery detection. Their approach combined convolutional layers with optimized pooling for local feature extraction. The model was tested on standard forgery datasets and achieved superior accuracy compared to traditional keypoint-based methods. However, the method showed some sensitivity to heavy post-processing such as compression and rotation.

Dhiman et al. [6] proposed a deep convolutional neural network tailored for forgery detection. The CNN was trained on a custom dataset of tampered and original images, and evaluation demonstrated improved precision and recall compared to handcrafted-feature baselines. Nonetheless, the model required extensive labeled training data, limiting scalability for unseen manipulations.

Diwan et al. [7] advanced this work through a multiscale detector for copy-move manipulation. Their system combined multiscale CNN filters to capture forgeries at varying patch sizes. Evaluated on multiple benchmark datasets, the method reported higher robustness in complex scenarios, including scaled or rotated duplicates. Yet, the computational overhead from multiscale processing limited real-time applicability.

Liu et al. [8] introduced CMFDFormer, a transformer-based architecture for copy-move forgery detection. The model leveraged self-attention to capture global dependencies and was coupled with continual learning to adapt to new forgery styles. Experiments on benchmark datasets showed that CMFDFormer outperformed CNN-only baselines in cross-dataset generalization, though training transformers demanded high computational resources.

Rao et al. [9] presented ResTran, which combined residual CNN layers with transformer blocks to capture long-distance relationships across the image. Using benchmark datasets, ResTran demonstrated improved sensitivity to spatially separated but semantically connected tampered regions. A limitation was that while accuracy improved, inference times were slower than purely CNN-based models.

Khalil et al. [10] explored transfer learning for digital image forgery detection. By fine-tuning pre-trained CNNs (such as VGG and ResNet) on forgery datasets, they reported strong improvements in accuracy and efficiency compared to training from scratch. The method

significantly reduced training costs but was still constrained by domain-specific biases in the pre-training datasets.

Chaitra and Reddy [11] proposed an optimized pre-trained deep learning model for multiple copy–move forgery detection. Their approach incorporated fine-tuning and optimization of CNN backbones, achieving high detection rates across multiple forgery types. The study showed improved generalization, though it required careful hyperparameter tuning to avoid overfitting.

Zhao et al. [12] developed SPA-Net, which enhanced CNN detection by integrating a span-partial structure and attention mechanism. Tested on multiple copy–move datasets, SPA-Net achieved state-of-the-art performance with improved robustness to post-processing operations. However, the model was relatively heavy in terms of parameter size, raising deployment concerns for low-resource environments.

Table 1: Comparative Analysis of Existing Forgery/Deepfake Detection Approaches

Authors (Year)	Model/Approach	Dataset Used	Key Results	Limitations
Hosny et al. (2022) [5]	Efficient CNN for copy–move detection	Benchmark forgery datasets	High accuracy, robust against simple copy–move attacks	Performance degrades with heavy compression and rotation
Dhiman et al. (2023) [6]	Deep CNN for image forgery detection	Custom tampered/original dataset	Improved precision & recall over handcrafted baselines	Requires large labeled training data; less effective on unseen manipulations
Diwan et al. (2024) [7]	Multiscale CNN detector	Multiple copy–move benchmarks	Strong robustness in complex manipulations; handles scaling/rotation	High computational cost limits real-time use
Liu et al. (2023) [8]	CMFDFormer (Transformer + continual learning)	Public forgery benchmarks	Outperforms CNN-only models; good cross-dataset generalization	Requires large computational resources; training overhead
Rao et al. (2023) [9]	ResTran (Residual CNN + Transformer)	Image forgery benchmarks	Captures long-distance dependencies; higher sensitivity to tampering	Slower inference than pure CNNs; deployment overhead
Khalil et al. (2023) [10]	Transfer learning with pre-trained CNNs (e.g., VGG, ResNet)	Forgery datasets	Achieved strong accuracy with less training data	Domain bias from pre-trained models affects generalization
Chaitra & Reddy (2023) [11]	Optimized pre-trained CNN for multiple forgery detection	Forgery datasets	High detection rates across multiple forgery types	Needs careful hyperparameter tuning; risk of overfitting
Zhao et al. (2023) [12]	SPA-Net (CNN + attention)	Copy–move forgery datasets	State-of-the-art accuracy; robust against post-processing	Large model size; high parameter count affects efficiency

Table I summarizes recent approaches in image forgery and deepfake detection, highlighting the diversity of models, datasets, and outcomes. CNN-based techniques [5]-[7] demonstrate strong capabilities in capturing local artifacts such as duplicated regions and texture inconsistencies, though they often struggle with heavy post-processing and dataset generalization. Transformer-driven architectures [8], [9] provide improved cross-dataset robustness by modeling long-range dependencies, but incur higher computational costs and slower inference. Hybrid and transfer learning methods [10], [11] effectively leverage pre-trained knowledge to reduce training requirements and improve adaptability, while attention-based models [12] enhance feature sensitivity and achieve state-of-the-art performance, albeit with large parameter footprints. Collectively, these studies reveal a trade-off between accuracy, generalization, and efficiency, reinforcing the need for a balanced hybrid framework—such as the proposed ResNet-50 feature extractor combined with VGG-16 classifier—that unites robust local feature learning with efficient classification for deepfake detection.

3 Research Gaps

Although significant progress has been made in image forgery and deepfake detection, several research challenges remain unresolved. The reviewed studies [5]-[12] highlight common limitations that motivate the need for improved methodologies:

1. **Limited Cross-Dataset Generalization**
CNN-based approaches [5]-[7] perform well on benchmark datasets but degrade substantially when applied to unseen manipulations. Models often overfit to dataset-specific artifacts, reducing their applicability to real-world deepfakes.
2. **Sensitivity to Post-Processing Operations**
Methods such as [5] and [12] show reduced accuracy when images are compressed, blurred, or resized. Since deepfake media is frequently shared online where compression is inevitable, robustness against such transformations is critical.
3. **High Computational Overhead**
Transformer-based and multiscale approaches [7]-[9] capture richer feature representations but are computationally expensive. This complexity makes them unsuitable for real-time detection or deployment in resource-constrained environments.
4. **Dependence on Extensive Training Data**
Deep CNN approaches [6] and transfer learning methods [10], [11] require large volumes of labeled forgery data for training or fine-tuning. The scarcity of balanced deepfake datasets limits the scalability of these solutions.
5. **Overfitting and Model Complexity**
Optimized pre-trained and attention-based models [11], [12] achieve strong accuracy but often rely on large architectures with high parameter counts, leading to overfitting and inefficiency in practical deployments.
6. **Insufficient Balance Between Local and Global Features**
Existing models tend to emphasize either local feature extraction (CNNs) or global context modeling (Transformers), but rarely both in an efficient manner. This imbalance limits their ability to detect subtle manipulations that require multi-level feature integration.

4 Proposed Methodology

To overcome the limitations identified in Section III, we propose a hybrid CNN-based framework that integrates ResNet-50 as a block-based feature extractor and VGG-16 as a transfer learning-based classifier. This combination is designed to exploit the residual learning capability of ResNet-50 for extracting robust features while leveraging the fine-tuned classification strength of VGG-16 for deepfake detection. Figure 1 shows the architecture of the proposed methodology.

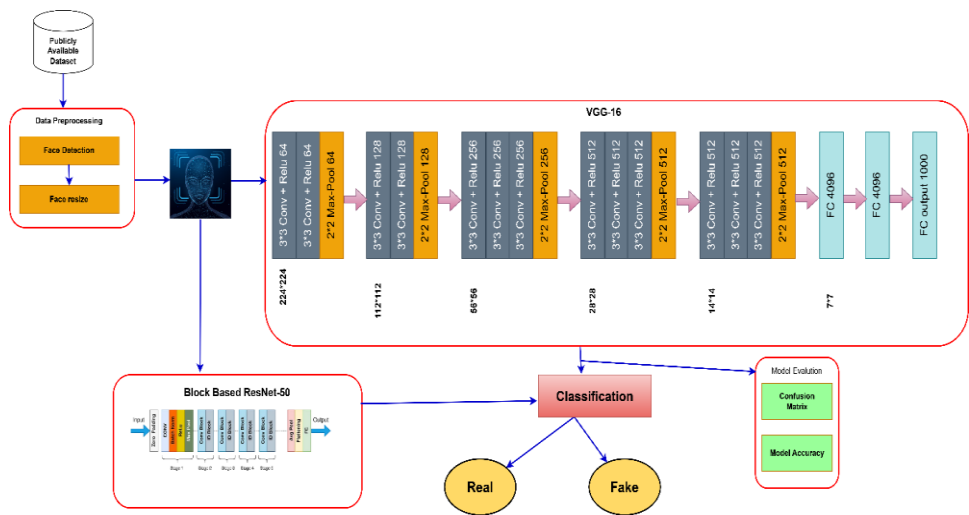


Fig 1: Architecture of proposed methodology

The methodology consists of several stages as described below.

A. Dataset Preparation

Experiments will be conducted on public datasets including Celeb-DF [13] and DeepFake Detection Challenge (DFDC) [14] which consist of real and fake images at scale. These datasets were selected for their variation in manipulation methods, human/identity diversity and image quality, and are challenging for generalization performance assessment. Each dataset will be split into 3 subsets: training (70%), validation (15%) and testing (15%) respectively. This split guarantees that the model is trained effectively and enables unbiased evaluation) through separate validation and test datasets.

B. Data Preprocessing

To normalize input data, all images are resized to 224×224, height and width, corresponding to the input shape of VGG-16. Normalizing pixel intensities between 0-1 will make the training process stable and faster in convergence. The ground-truth labels will be encoded in binary format, with “0” as the true images and “1” as the deepfakes. Besides, a number of data augmentation operations (random rotate, horizontal flip, scale, and brightness etc.) will be performed. These augmentations induce more diversity to the training sample set, preventing overfitting, and furthermore, make the model more robust to standard post-processing operations as compression or blurring.

C. Feature Extraction with ResNet-50

ResNet-50 will be working as the primary feature extractor due to its deep residual architecture and proven ability to capture hierarchical image representations. It is the residual connections in ResNet-50 that overcome the vanishing gradient problem and make it possible to train deep networks. We will use ResNet-50 pretrained weights on ImageNet to utilize generalized feature representations in the first step. While freezing its layers in the first training stage to keep the knowledge of weights and reduce computational load. The upper levels of then ResNet-50 are fine-tuned in the later stages on deepfake data so that the network focuses on learning features that are identifying manipulation-specific features.

D. Classification with VGG-16

ResNet-50 is used to extract the feature and some of the features are used as input of a fine-tuned VGG-16 to classify. VGG-16 is selected because it has a simple but effective layout, with 13 convolutional layers and 3 dense layers. The bottom convolutional layers, which detect simple visual patterns, such as edges and textures, are frozen, and only the top convolutional layers and the fully connected layers are fine-tuned using the deepfake datasets. This scheme of transfer learning allows the model to maintain learn robust generic features and at the same time, adapt to match the task-specific demands of deepfake detection. A binary softmax classifier is then added at the end to classify images as either real or fake.

E. Training Strategy

The learning process will be performed in two steps. In the first phase, we trained and fine-tuned the VGG-16 classifier with features from the superficial ResNet-50. In the 2DoF, we will perform selective fine-tuning in the upper layers of ResNet-50 together with VGG-16 to further specialize this system in the task of deepfake recognition. Binary cross-entropy loss will be minimized using the Adam optimizer. Early stopping and updating of the learning rate will be adopted to avoid overfitting and to achieve stable convergence.

F. Evaluation Metrics

Cross validation will be chosen as the validation method, and conforming to practice in binary classification problems, performance will be evaluated with standard metrics. Accuracy is a metric that will quantify the general level of rightness of the predictions, while precision and recall will inform on trustworthiness and inclusion of labeled content. The F1-score (for balancing precision and recall) will be computed, and the AUC-ROC curve will be constructed to visualize classifier performance on varying thresholds. A combination of these metrics will allow us to have a holistic view of the model performance as well as its generalization capacity to multiple data domains.

5 Challenges in Deepfake Image Detection

Despite the advancements in detection of deepfake and image forgery, a number of challenges are still hinder reliable and scalable deployment of detection systems.

There exist several challenges, e.g., dataset bias and generalization. In practice, most detection models are trained and tested on dedicated datasets, such as FaceForensics++ and Celeb-DF. Unfortunately, such data sets are often not realistic in that they do not cover the diversity of manipulations that are typically encountered in the real world. As a result, models trained on a particular dataset tend to perform poorly on previously unseen manipulations or low-quality videos [15].

A related challenge is the hypothesize-verify loop and the robustness against adversarial attacks. Recent research has demonstrated that deepfake detectors can be deceived by well-crafted perturbations that are not obvious to the human eye but trick machine systems. This

weakness not only casts doubt on the trustworthiness of current models, but may also provide opportunities for evading security-critical systems for adversaries [16]. The problem of scalability and computational complexity also make the approaches impractical. Current cutting-edge detection approaches, especially transformer-based models, pose high demand of computational resources and memory usage, which greatly impedes their wide adaption and utilization for real-time applications like social media monitoring and forensics [17].

In addition, real-world deployment challenges arise from post-processing and compression. Deepfake content shared across platforms often undergoes resizing, format conversion, or lossy compression, all of which significantly reduce detection accuracy. While augmentation techniques partially address this issue, achieving consistent robustness across diverse post-processing conditions remains unsolved [18].

Finally, there is an increasing demand for multimodal detection frameworks that combine visual, audio, and textual cues. Current research primarily focuses on image or video modalities, but multimodal approaches are likely to provide more reliable verification in adversarial environments. However, integrating multimodal features introduces complexities in data alignment and model training, which remain open research challenges [19].

Collectively, these challenges highlight the need for hybrid, efficient, and adaptive detection frameworks that balance accuracy, robustness, and real-world deployability. The proposed ResNet-50 and VGG-16 hybrid model is designed with these concerns in mind, aiming to reduce dataset bias, improve feature robustness, and maintain computational efficiency.

6 Conclusion and Future Scope

Deepfake detection has become one of the most crucial problems in multimedia forensics owing to the rapid advancement of generative models, which can be abused for spreading fake news, identity theft and social manipulation. In this paper, we surveyed the state-of-the-art methods of deepfake and image forgery detection, emphasizing the advantages of CNN-based, transformer-based and a hybrid architecture. The study shows persistent research gaps, namely cross-dataset generalization, resistance to post-processing and adversarial perturbations, and the trade-off between accuracy and computational cost.

To solve these problems, this paper presents a hybrid system that employs ResNet-50 for strong block-based features extraction and VGG-16 for fine-tuned classification. This structure is developed in order to improve the accuracy of detection and to keep scalability for practical application. With support of the merits brought by residual learning and transfer learning, the framework offers a promising direction to address limitations found in existing works.

Future work shall focus on cross-dataset learning, and continual adaptation to further mitigate the reliance on limited training data and enhancing generalization across manipulation types. Adversarial robust training methods will be critical to protect models from intentionally crafted attacks to evade detection. In addition, multimodal networks that incorporate visual, audio, and textual signals, may give rise to a more reliable system in practice. Despite the difficulty of deepfake detection, hybrid models such as the ResNet-50 and VGG-16 framework are an important step toward developing robust, general-purpose and efficient detection systems capable of addressing the evolving threats of digital forgeries.

References

1. P. Korshunov and S. Marcel, "Deepfakes: a new threat to face recognition? Assessment and detection," *arXiv:1812.08685*, 2019.

2. A. Rossler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner, "FaceForensics++: Learning to detect manipulated facial images," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2019, pp. 1–11.
3. D. Afchar, V. Nozick, J. Yamagishi, and I. Echizen, "MesoNet: A compact facial video forgery detection network," in *Proc. IEEE Int. Workshop Inf. Forensics Secur. (WIFS)*, 2018, pp. 1–7.
4. T. T. Nguyen, C. M. Nguyen, D. T. Nguyen, D. T. Nguyen, and S. Nahavandi, "Deep learning for deepfakes creation and detection: A survey," *arXiv:1909.11573*, 2019.
5. K. M. Hosny, A. M. Mortda, M. M. Fouda, and N. A. Lashin, "An efficient CNN model to detect copy-move image forgery," *IEEE Access*, vol. 10, pp. 48622–48632, 2022, doi: 10.1109/ACCESS.2022.3172273.
6. N. Dhiman, H. Singh, and A. Thakur, "An efficient approach for image forgery detection using deep convolutional neural network," in *Proc. Int. Conf. Electr., Electron., Commun. Comput. (ELEXCOM)*, Roorkee, India, 2023, pp. 1–5, doi: 10.1109/ELEXCOM58812.2023.10370304.
7. A. Diwan, R. Mahadeva, and V. Gupta, "Advancing copy-move manipulation detection in complex image scenarios through multiscale detector," *IEEE Access*, vol. 12, pp. 64736–64753, 2024, doi: 10.1109/ACCESS.2024.3397466.
8. Y. Liu, C. Xia, S. Xiao, Q. Guan, W. Dong, Y. Zhang, and N. H. Yu, "CMFDFormer: Transformer-based copy-move forgery detection with continual learning," *arXiv:2311.13263*, 2023.
9. J. Rao, S. Teerakanok, and T. Uehara, "ResTran: Long distance relationship on image forgery detection," *IEEE Access*, vol. 11, pp. 120492–120501, 2023, doi: 10.1109/ACCESS.2023.3327761.
10. A. H. Khalil, A. Z. Ghalwash, H. A.-G. Elsayed, G. I. Salama, and H. A. Ghalwash, "Enhancing digital image forgery detection using transfer learning," *IEEE Access*, vol. 11, pp. 91583–91594, 2023, doi: 10.1109/ACCESS.2023.3307357.
11. B. Chaitra and P. V. B. Reddy, "An approach for copy-move image multiple forgery detection based on an optimized pre-trained deep learning model," *Knowl.-Based Syst.*, vol. 269, 110508, 2023, doi: 10.1016/j.knosys.2023.110508.
12. K. Zhao, X. Yuan, Z. Xie, Y. Xiang, G. Huang, and L. Feng, "SPA-Net: A deep learning approach enhanced using a span-partial structure and attention mechanism for image copy-move forgery detection," *Sensors*, vol. 23, no. 6430, 2023, doi: 10.3390/s23146430.
13. Y. Li, X. Yang, P. Sun, H. Qi, and S. Lyu, "Celeb-DF: A large-scale challenging dataset for deepfake forensics," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3207–3216.
14. B. Dolhansky, R. Howes, B. Pflaum, N. Baram, and C. C. Ferrer, "The Deepfake Detection Challenge dataset," *arXiv preprint arXiv:2006.07397*, 2020.
15. H. Haliassos, J. Vougioukas, S. Petridis, and M. Pantic, "Lips Don't Lie: A Generalisable and Robust Approach to Face Forgery Detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2021, pp. 5039–5049.
16. M. Neekhara, B. Dolhansky, R. Howes, M. Wang, J. Bitton, and C. C. Ferrer, "Adversarial threats to deepfake detection: A practical perspective," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, 2021, pp. 3347–3356.
17. S. Tariq, S. Lee, H. Kim, and S. Woo, "Detecting both machine and human created fake face images in the wild," in *Proc. 2nd Int. Workshop Multimedia Privacy and Security (MPS '18)*, 2018, pp. 81–87.

18. Y. Jeon, J. Heo, H. Lee, and J. Nam, “FDFtNet: Facing off fake images using fake detection fine-tuning network,” in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, 2020, pp. 2464–2473.
19. P. Korshunov and S. Marcel, “VoxCeleb2 for learning deepfake detection from audio-visual cues,” in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2020, pp. 2967–2971.